

Healthy teams achieve better
results.

Employee Wellness Management Analytics

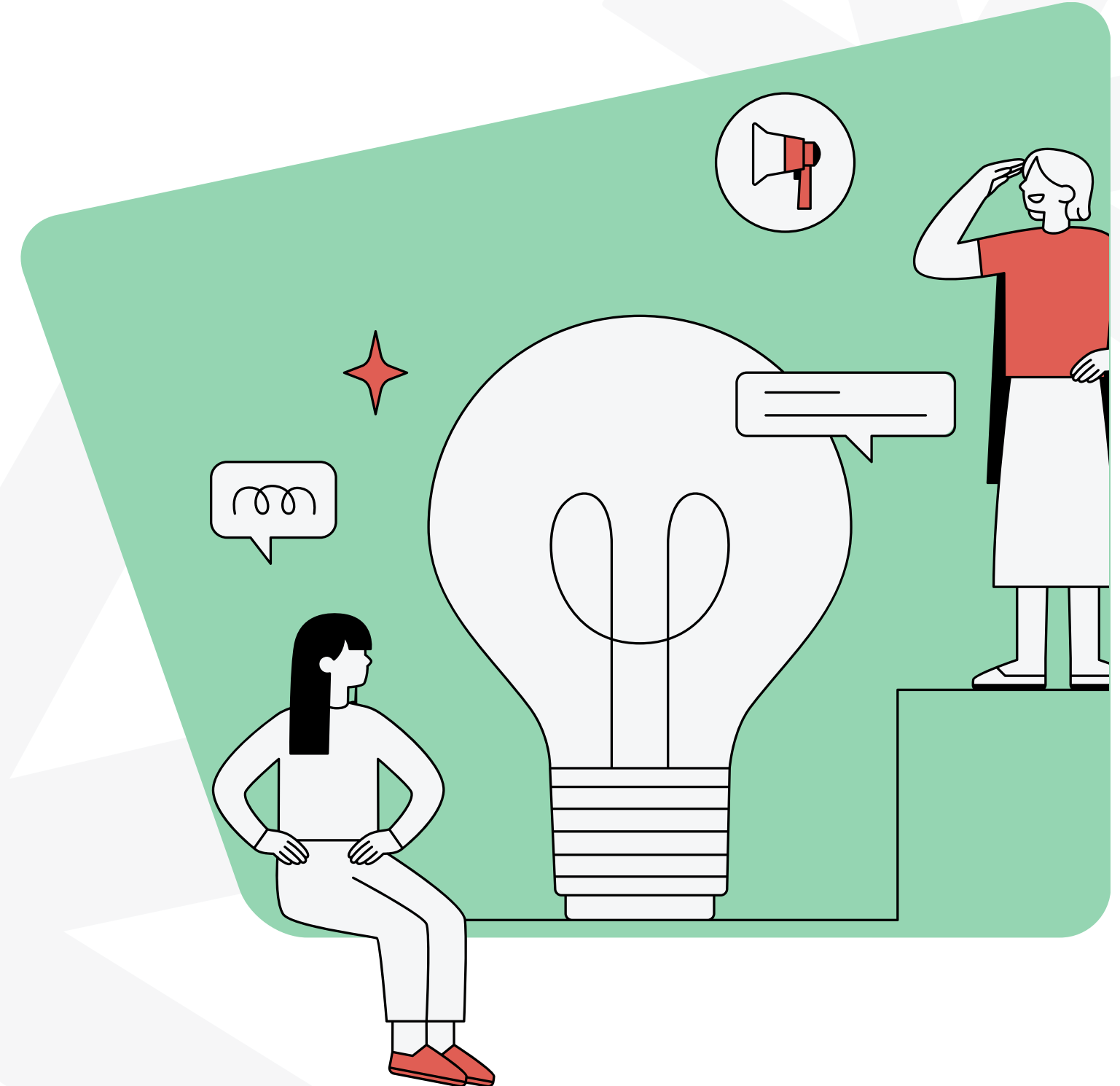
Support Employee's Mental Health and
Sentiment Remotely

Presented By:

Team 4



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Sudip Madhu**



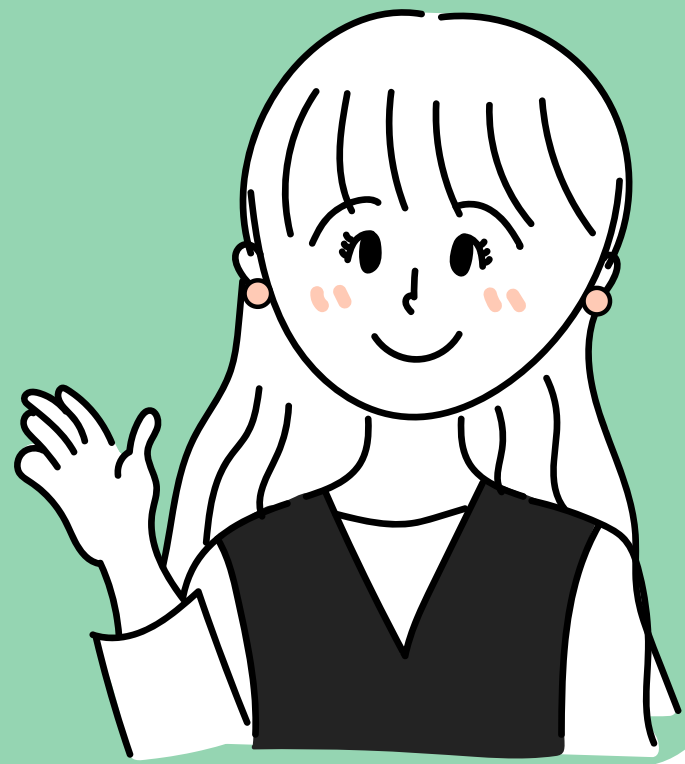
Problem Statement



→ Current System's Limitations and Issues

- Lack of a secure and centralized login and wellness management system.
- Difficulty in managing user/admin access and employee wellness records.
- Limited real-time analysis of workforce performance and health trends.
- Need for instant wellness guidance through an intelligent chatbot with natural language response.





Introduction

→ **Knowing about the Project**

- Employee wellness is essential for strong productivity and steady growth. The system helps manage health data and spot risks like stress while offering personal wellness tips.
- Employee wellness plays a big role in performance and growth. This system handles health data, predicts risks, and gives each person wellness ideas that support their wellbeing.
- Supporting employee wellness leads to better focus and stronger results. The system works with wellness data, identifies stress risks, and shares recommendations and the sentiment to help everyone stay healthy.

Project Objective



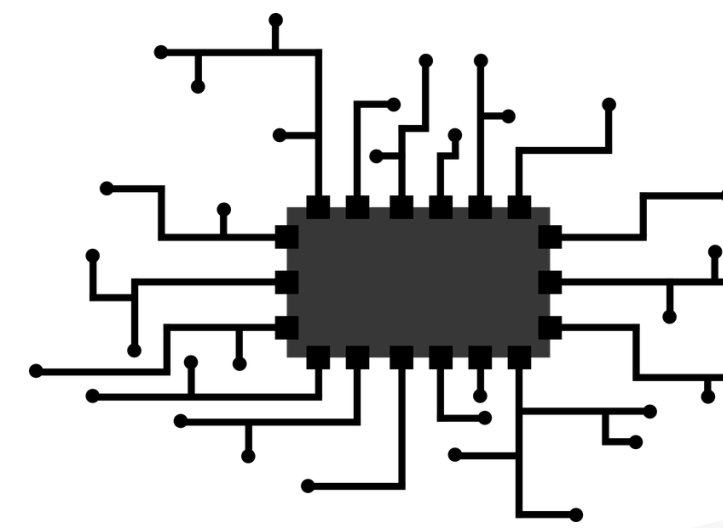
→ Objectives of the Project

1. Personalized Employee Dashboards: Tracks vitals, sets goals, and delivers custom AI insights.
2. Proactive Admin Analytics: Enables predictive risk modeling, sentiment tracking, and key KPIs.
3. 24/7 AI Wellness Assistant: Provides instant, contextual advice on diet, stress, and exercise.
4. Secure & Scalable Infrastructure: Ensures data privacy, role-based access, and centralized analytics.
5. Automated Admin & Emergency Response: Streamlines claims, scheduling, and instant SOS alerts.



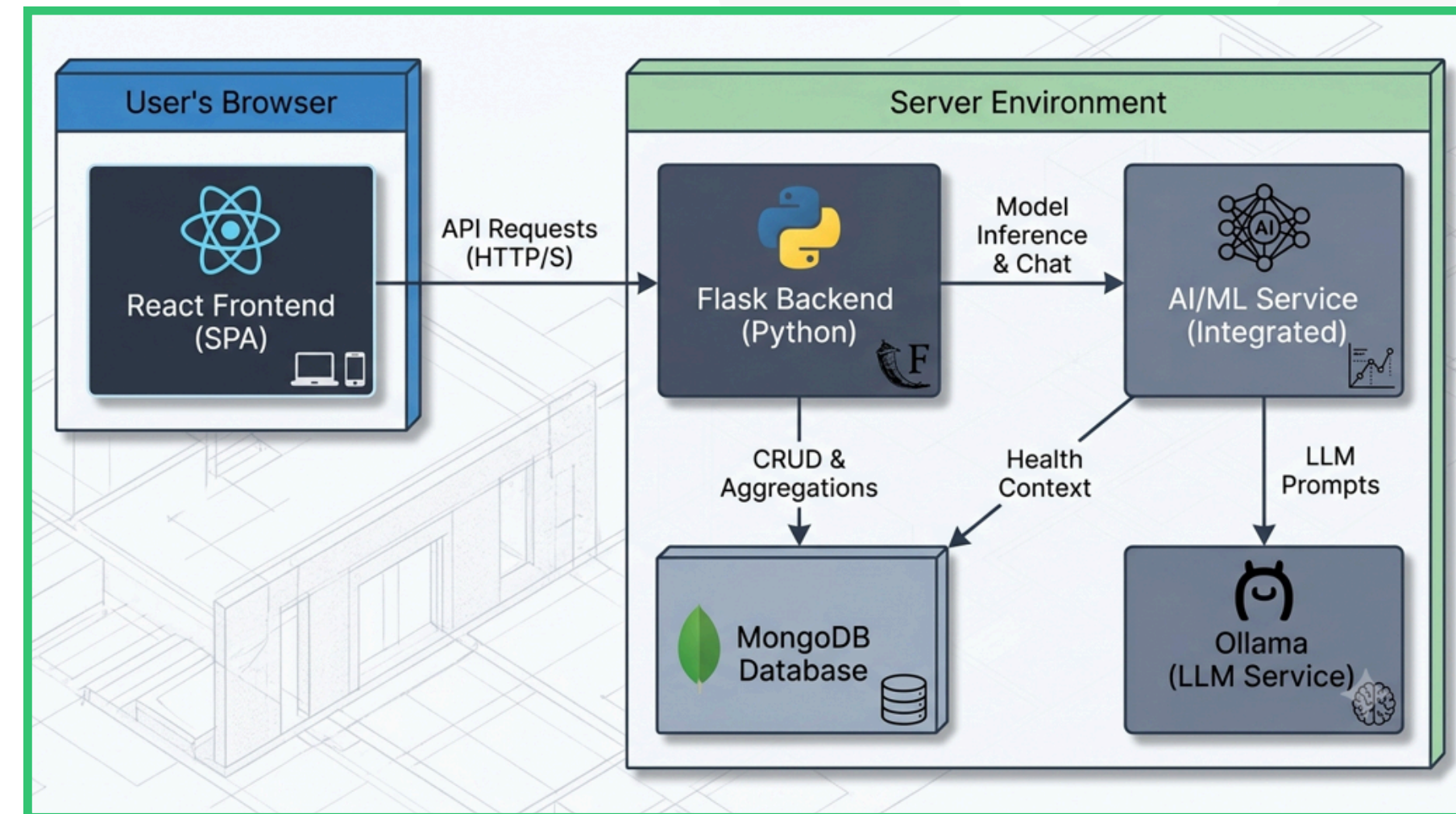


System Architecture

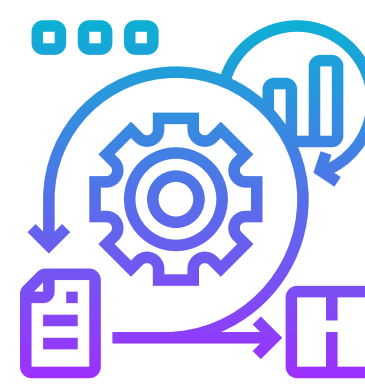


→ Explaining the structure and flow of the system.

- The application is built on a modern, decoupled architecture, commonly known as a Backend for Frontend (BFF) pattern.
- This means the React frontend is a distinct application that communicates with a dedicated Flask backend via a REST API
- Microservice-Ready AI Integration: Isolates LLM interactions and machine learning model inference into modular background services to maintain high responsiveness and system efficiency.
- Stateless Token Authentication: Utilizes secure JWT authorization across API requests, ensuring strict role-based access control and seamless horizontal scaling.

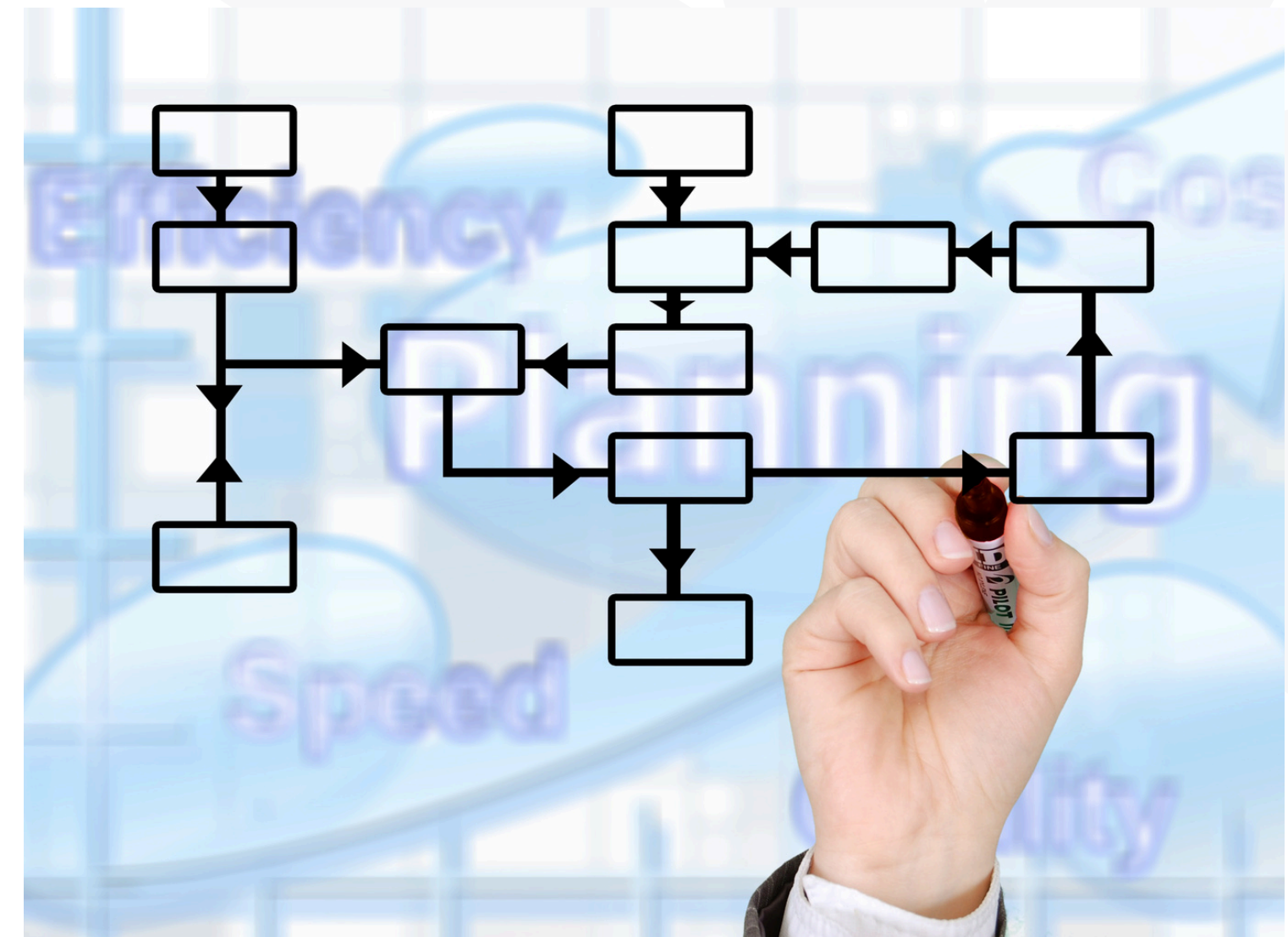


Methodology

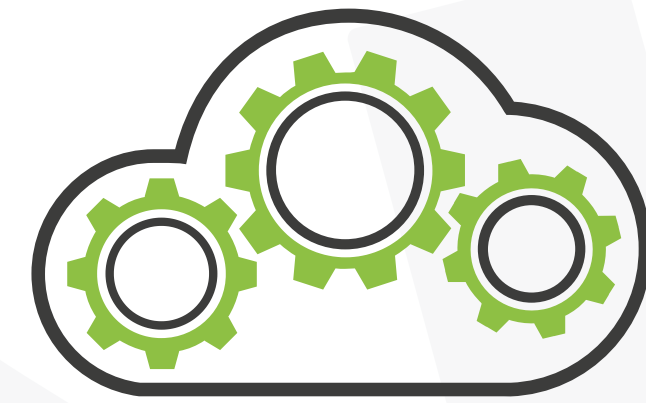


→ How the system was developed and works step-by-step.

- The system was developed in a modular way, starting with secure user authentication and role-based access for employees and admins.
- After that, the frontend, backend, and database were connected to manage wellness records, analytics, and user interactions smoothly.
- Finally, AI-based features such as risk prediction, recommendations, sentiment analysis, performance analytics, and chatbot support were added to make the platform smarter and more useful.

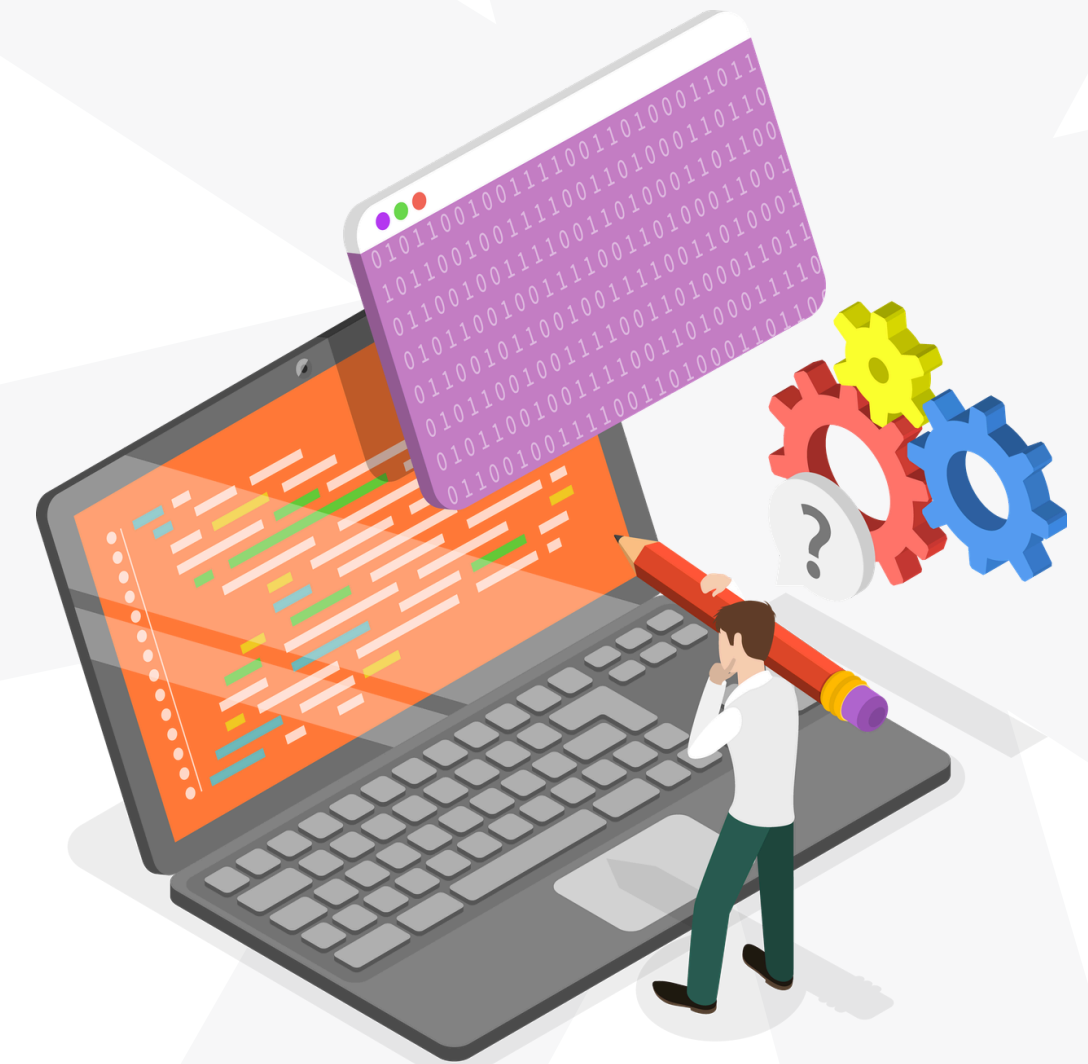


Software Specification



→ Software Requirements to run on local machine

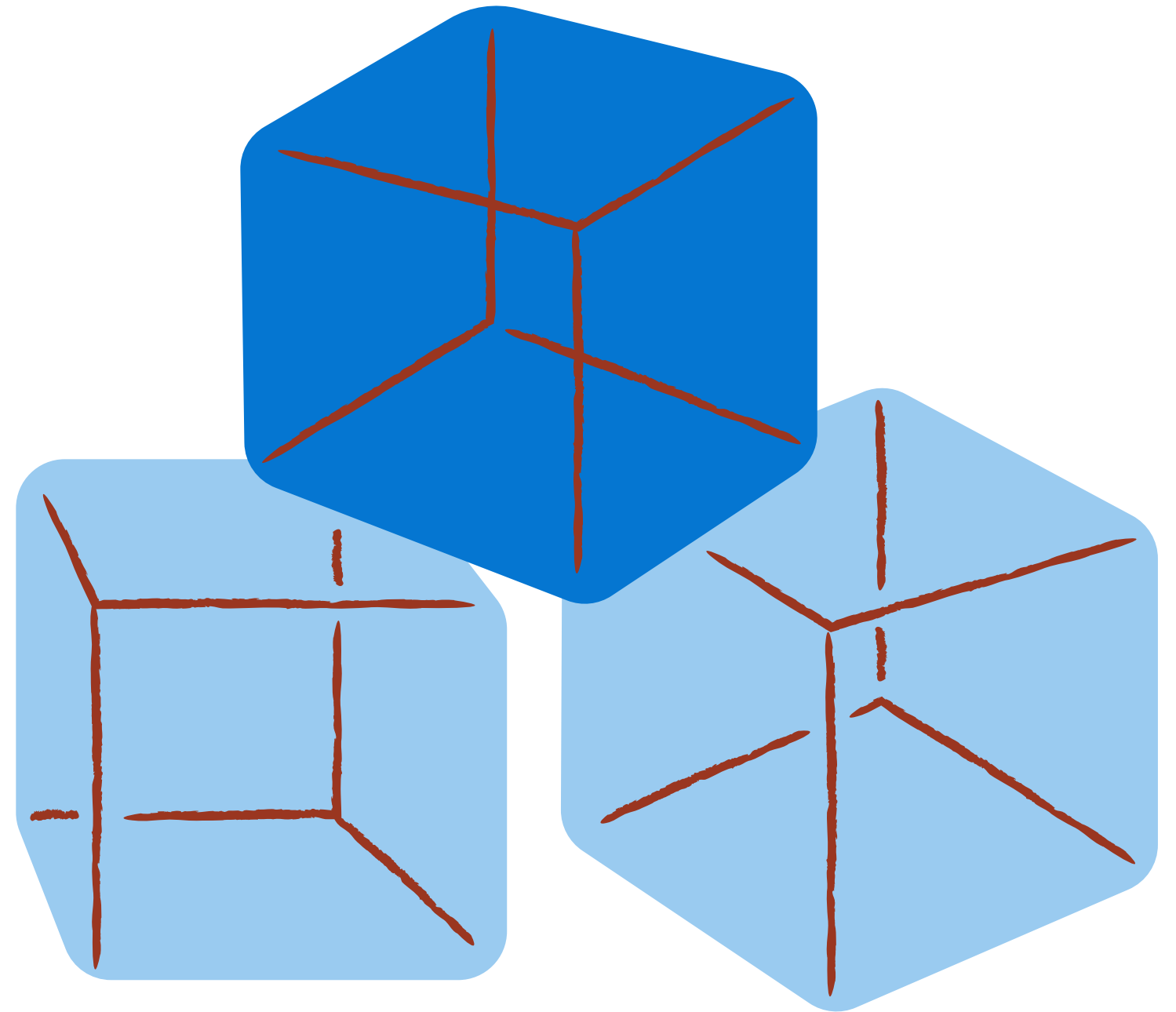
- Flask API with Python and NodeJs.
- Database: MongoDB.
- Authentication: JWT and bcrypt for secure login and password handling.
- Tools/Libraries: React Router, Flask-CORS, PyMongo, and ML libraries for wellness analytics.



Core



Modules





Login & SignUp Module



SignUp and SignIn



- User login with email and password.
- User registration with required fields.
- Remember me option in login form.
- Forgot password flow with email, OTP, and reset password steps in the frontend prototype.
- Role-based redirect to user or admin dashboard after authentication.

Create your account

Deploy wellness metrics and trackers for your team

FULL NAME

 John Doe

WORK EMAIL

 john.doe@company.com

PASSWORD

 At least 6 characters 

CONFIRM PASSWORD

 Match password 

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#).

Create Account →

Already have an account? [Sign In](#)



Employee Sign In

Use your registered employee credentials.

SELECT YOUR ROLE

 Employee

 Admin

EMPLOYEE ID

EMP101

EMAIL ADDRESS

 rishitranjan131@gmail.com

PASSWORD

[Forgot password?](#)

☐ Remember me

Sign In →

Don't have an account? [Create account](#)

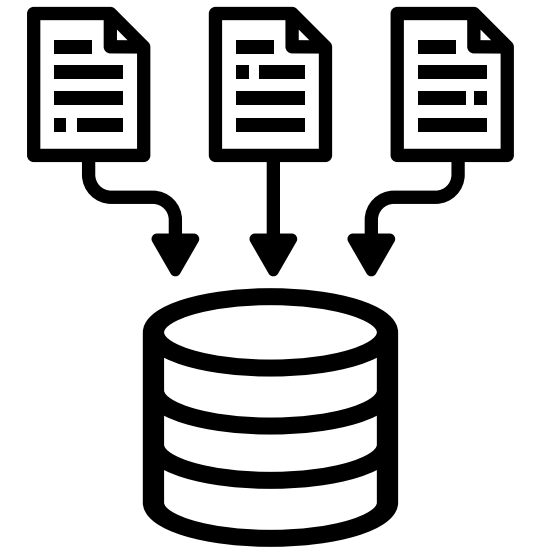
LOGIN!

Database

➔ Database Management of the Employee Data

The backend connects to MongoDB using PyMongo.

User-related data is stored in collections including admin and employees in JSON - like format.



Cluster0 > employee_wellness_analytics				View monitoring		Visualize your data		+ Create collection	
Collection name ⌵⌴	Properties ⌵⌴	Storage size ⌵⌴	Data size ⌵⌴	Documents ⌵⌴	Avg. document size ⌵⌴	Indexes ⌵⌴	Total index size		
admin	-	36.86 kB	245.00 B	1	245.00 B	1	36.86 kB		
daily_habits	-	36.86 kB	301.00 B	3	100.00 B	1	36.86 kB		
employee_feedback	-	4.10 kB	0 B	0	0 B	1	4.10 kB		
health_history	-	36.86 kB	947.00 B	3	315.00 B	1	36.86 kB		
health_records	-	36.86 kB	2.04 kB	3	681.00 B	1	36.86 kB		
mental_health_logs	-	36.86 kB	759.00 B	5	151.00 B	1	36.86 kB		
notifications	-	36.86 kB	265.00 B	1	265.00 B	1	36.86 kB		
password_reset_requests	-	32.77 kB	0 B	0	0 B	1	32.77 kB		
sentiment_pulses	-	4.10 kB	0 B	0	0 B	1	4.10 kB		
users	-	36.86 kB	792.00 B	3	264.00 B	1	36.86 kB		

Health Records Collection

- ➔ Collecting the Health Record
- BMI and medical reports.
 - Exercise and sleep tracking.
 - Attendance and assessments.
 - Data preprocessing and validation.

Employee Health Data Management

Database logs for tracking key metrics including BMI, medical stats, sleep, and lifestyle routines.

Analytics Active

Search employee or ID...

All Departments

+ Add Employee's Health Profile

RI Rishit Ranjan

EMP101

Dept: Engineering Age: 22

BMI: 23 Normal BP: 120/80

Ex (hrs/wk): 3 Sleep (hrs/nt): 8

Stress: Medium Glucose: 90

Condition: No major condition

Last Sync: Aug 2 2026 10:30 PM Fair

SU Sudip Madhu

EMP102

Dept: Engineering Age: 19

BMI: 23.5 Normal BP: 120/80

Ex (hrs/wk): 3 Sleep (hrs/nt): 6

Stress: Medium Glucose: 78

Condition: No major condition

Last Sync: Jul 30 2026 5:30 AM Good

SH Shiv

EMP103

Dept: Engineering Age: 21

BMI: 18.8 Normal BP: 120/80

Ex (hrs/wk): 0 Sleep (hrs/nt): 7.5

Stress: High Glucose: 90

Condition: No major condition

Last Sync: Jul 29 2026 5:30 AM Needs Attention

SI Simran Hakim

EMP105

Dept: Engineering Age: 18

BMI: 24.2 Normal BP: 120/80

Ex (hrs/wk): 3.5 Sleep (hrs/nt): 3.5

Stress: Medium Glucose: 90

Condition: No major condition

Last Sync: Jul 28 2026 5:30 AM Fair

SA Sanjana K Teggi

EMP104

Dept: Engineering Age: 20

Wellness Risk Prediction

→ Predicting the Risk

- Predicts stress and burnout.
- Detects obesity and hypertension risk.
- Uses historical and real-time data.
- ML: Decision Tree, Random Forest



MODULE 2 OF 5

Wellness Risk Prediction

Machine learning assessments predicting health risks, cardiovascular issues, or stress burnout.

HIGH SEVERITY



0 Score $\geq 70\%$

Critical risk indicators. Immediate clinical review or stress PTO mandated.

MODERATE SEVERITY



0 Score 45-69%

Elevated stress triggers. Guided meditation and ergonomic desk updates advised.

LOW SEVERITY



3 Score $< 45\%$

Healthy baseline. Maintain current lifestyle routines and claim fitness rewards.

Filter risk records by clinical severity:

All Risks (3)

High (0)

Moderate (0)

Low (3)

Rishit Ranjan

LOW SEVERITY
Category

EMP101

Risk Intensity Index: 42%

TRIGGERS DETECTED

Vitals check within ideal levels

PRESCRIBED ACTION

Low risk profile. Maintain current healthy routines and continue periodic monitoring.

Shiv

LOW SEVERITY
Category

EMP103

Risk Intensity Index: 41%

TRIGGERS DETECTED

Vitals check within ideal levels

PRESCRIBED ACTION

Low risk profile. Maintain current healthy routines and continue periodic monitoring.

Sudip Madhu

LOW SEVERITY
Category

EMP102

Risk Intensity Index: 38%

TRIGGERS DETECTED

Vitals check within ideal levels

PRESCRIBED ACTION

Low risk profile. Maintain current healthy routines and continue periodic monitoring.

Personalized Wellness Recommendation



Wellness Recommendation System

Tailored, evidence-based fitness routines, diet schedules, and mental wellbeing recommendations.

🔍 Search by employee name or ID...

Rishit Ranjan
EMP101

Low Risk



MENTAL WELLNESS

Guided Meditation Routine

10-15 minutes of guided mindfulness daily.

REASONS:

- Matches low-risk wellness profile



LIFESTYLE

Sleep Hygiene Program

Consistent bedtime routines and screen-free zones.

REASONS:

- Matches low-risk wellness profile



DIET

High-Protein Lean Macro Guide

Nutritional breakdown focusing on preservation of muscle tissue.

REASONS:

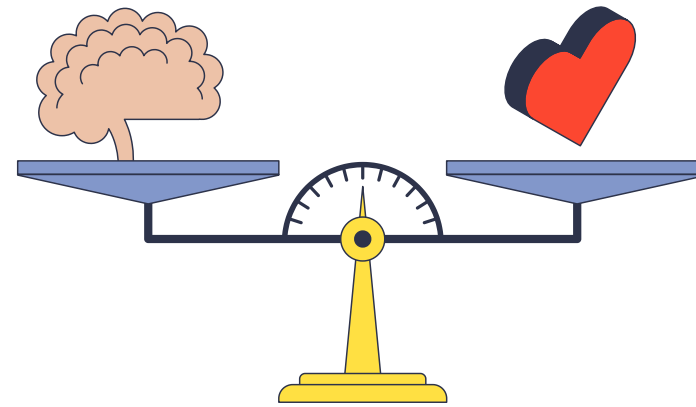
- Matches low-risk wellness profile
- Exercise can be improved further



Employee Wellness Recommendation

- Personalized fitness plans
- Diet and lifestyle suggestions
- Mental wellness and yoga activities
- AI: collaborative filtering, content-based filtering, Generative AI

Mental Health & Sentiment Analysis



Mental Health & Sentiment Analytics

NLP-driven individual stress analytics collected through fully anonymized feedback pulse-checks.

Analytics Active

Individual Employee Sentiment

Rishit Ranjan

EMP101 - Engineering

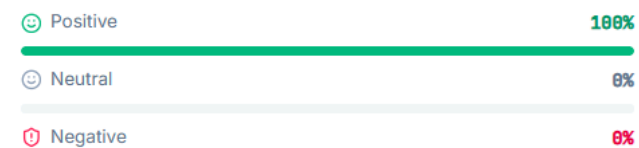
Medium

6.6/10

STRESS SCORE

Latest Mood: Neutral

Fair



RECENT FEEDBACK

i feel good working here

Sudip Madhu

EMP102 - Engineering

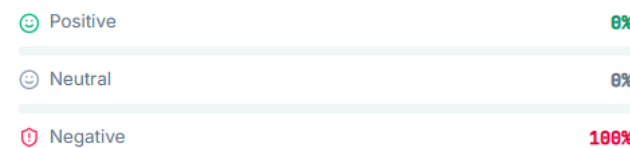
Medium

5.5/10

STRESS SCORE

Latest Mood: Neutral

Good



RECENT FEEDBACK

Shiv

EMP103 - Engineering

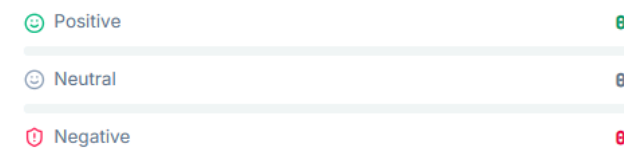
High

6/10

STRESS SCORE

Latest Mood: Neutral

Needs Attention



RECENT FEEDBACK

No recent feedback logs.

Simran Hakim

EMP105 - Engineering

Medium

5/10

STRESS SCORE

Latest Mood: Neutral

Fair

Sanjana K Teggi

EMP104 - Engineering

Medium

5/10

STRESS SCORE

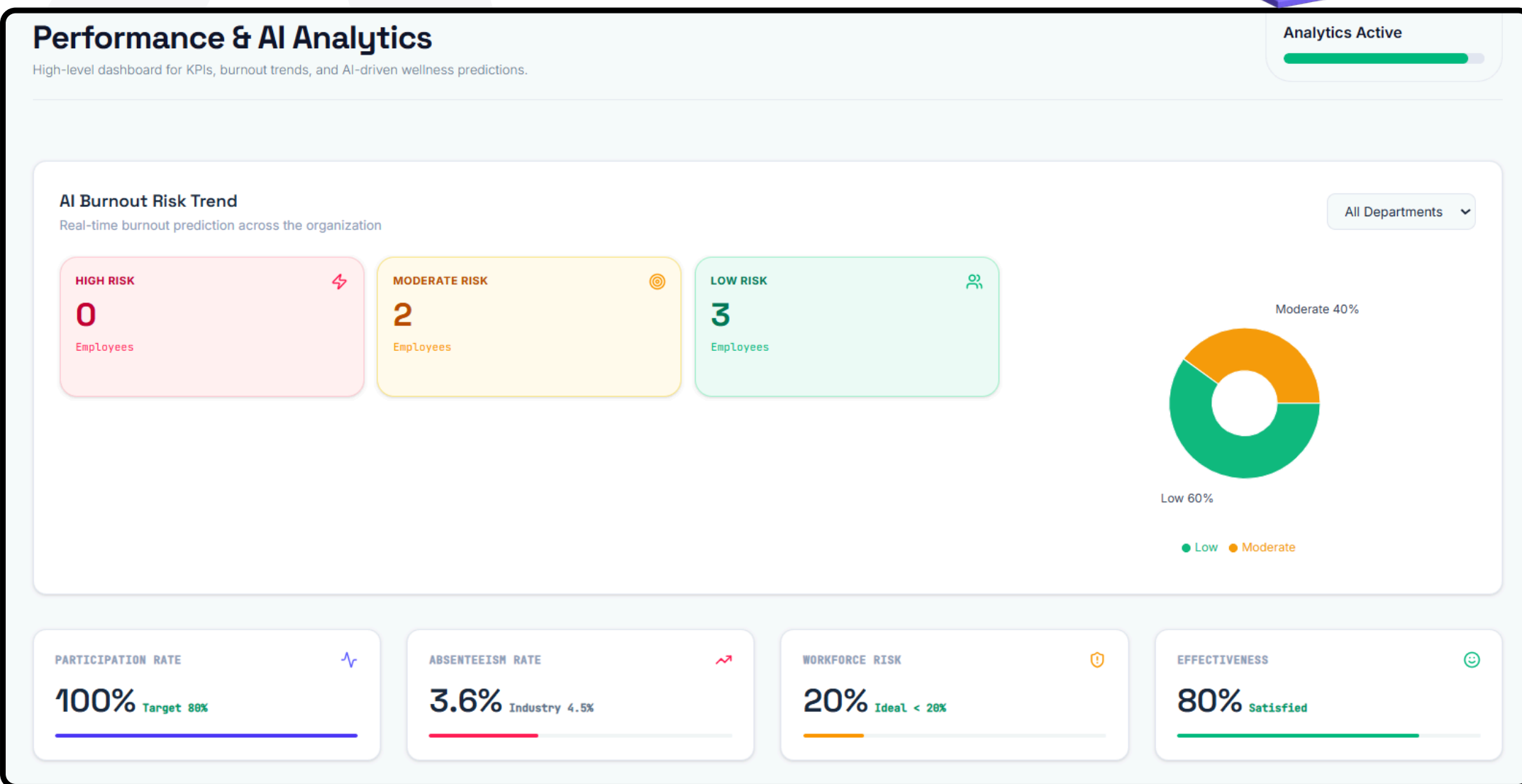
Latest Mood: Neutral

Good

➔ Sentiment and Mental Health

- Analyzes surveys and feedback.
- Can use emails with consent and chatbot conversations
- Detects stress, anxiety, burnout.
- NLP, sentiment analysis, emotion detection

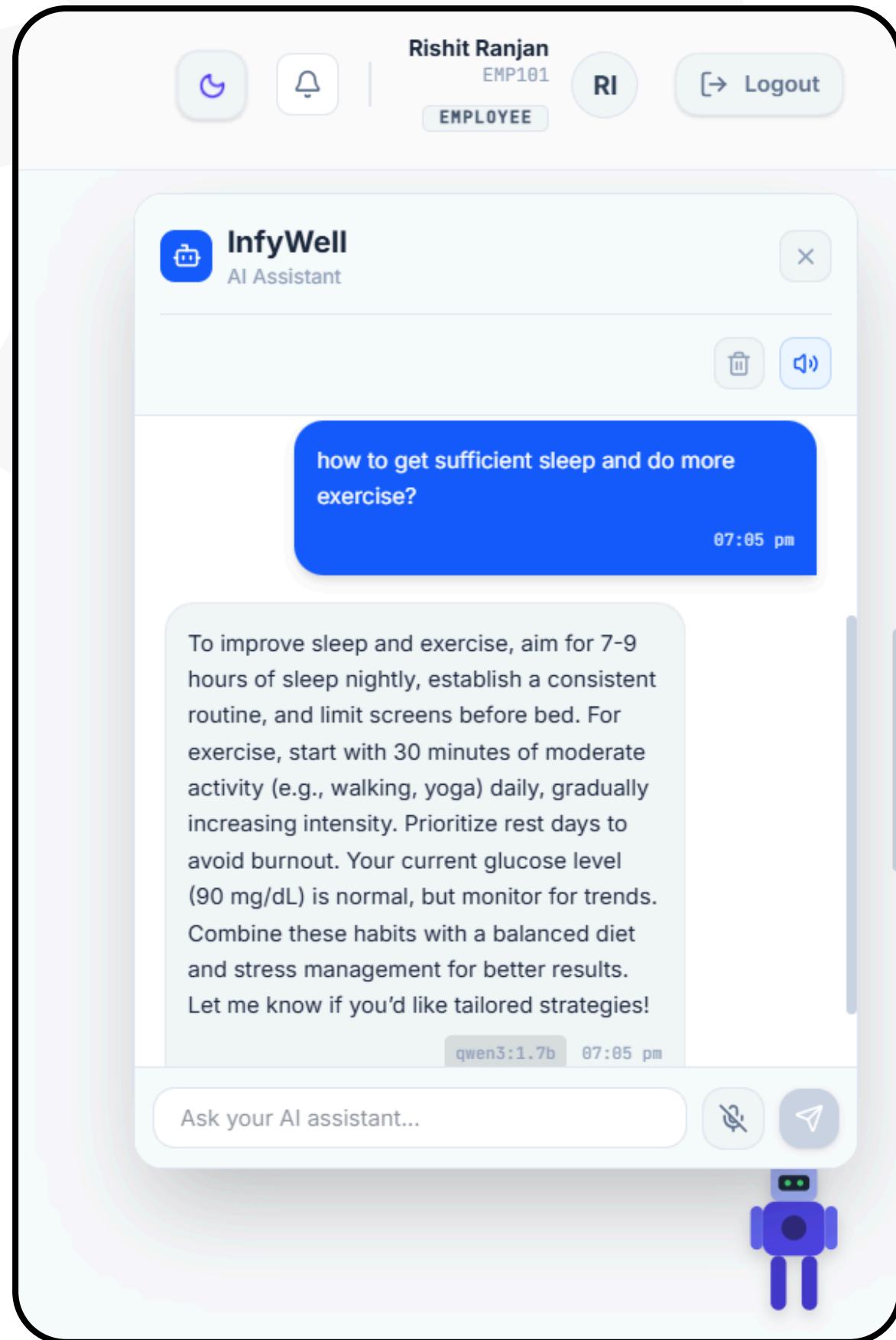
Performance & AI Analysis



Employee Performance Metrics

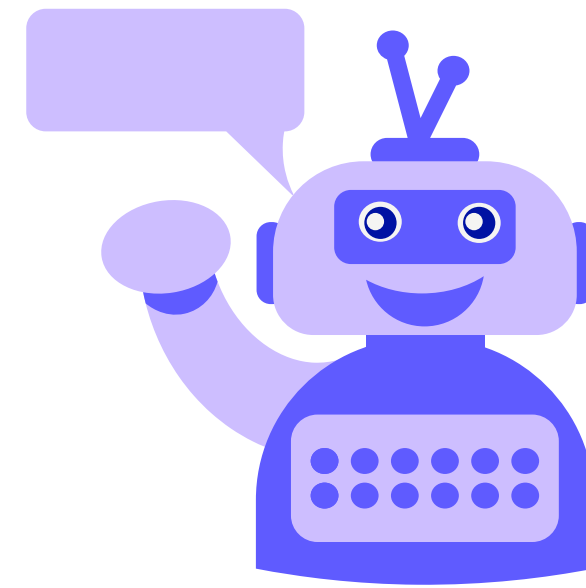
- Tracks absenteeism and workforce health risk.
- Measures wellness program effectiveness.
- Helps admin monitor employee performance trends.

Chatbot Module



Chatbot Support and Assistance

- AI-powered wellness assistant for employees.
- Answers questions on sleep, diet, exercise, stress, and burnout.
- Provides interactive chat support inside the user dashboard.



Additional



Features



Admin Dashboard



 Notification Center

 Compose Notification

Title (e.g. Annual Health Camp)

General

Message...

Broadcast to all employees

Send

 Sent Notifications

No notifications sent yet.

Checkups, SOS & Expenses

Oversee employee checkup scheduling, SOS alerts, and expense claims.

All Check-up Appointments

Employee	Type	Date	Status	Actions
EMP163 (Shiv)	Annual Health Check-up	2026-07-23	Completed	Completed

Emergency SOS Alerts

Simran Hakim (EMP105) 27/7/2026, 8:31:38 pm Contact: — Phone: — Blood Group: — Allergies: —	Resolved
Sudip Madhu (EMP102) 27/7/2026, 7:21:08 pm Contact: — Phone: — Blood Group: — Allergies: —	Resolved
Shiv (EMP103) 24/7/2026, 8:42:57 pm Contact: — Phone: — Blood Group: — Allergies: —	Resolved
Shiv (EMP103) 22/7/2026, 8:43:50 pm	Resolved

➔ Added Features for Analytics

- Added wellness-focused extension modules.
- Send announcements across all employees or each individually.

Admin Dashboard



System Settings

Manage application-wide settings and configurations.

Risk Thresholds

Define the score boundaries for wellness risk categories.

High Risk Threshold (%)

70

Medium Risk Threshold (%)

45

Notifications

Manage how the system communicates with users.

Enable Email Notifications

☒

Enable SMS Notifications

☐

Data & Privacy

Manage data retention and user privacy settings.

Data Retention (Days)

365

Anonymize Sentiment Data

☒

Save All Settings

SYSTEM VITALS

Analytics Active

EXTENSION MODULE 6

Insurance Management

Manage employee insurance policies, claims, and coverage oversight.

+ Assign / Update Policy

EMPLOYEE	PROVIDER	POLICY NO.	COVERAGE (₹)	EXPIRY
Select...	Star Health			dd-mm-yyyy

 Save Policy

☒ Pending Claims (0)

No pending claims.

☒ All Policies

No policies assigned yet.



Added Features for Analytics

- Track employee claims and policy status.
- System Settings for Managing application-wide settings and configurations.

Employee Dashboard



AI Wellness Assistant

AI-powered daily wellness insights, personalized routines, and intelligent coaching.

SLEEP SCORE

88% / 100

STRESS INDEX

42% / 100

ACTIVITY LEVEL

Active

Based on weekly fitness logs

NUTRITION

Balanced

Diet quality index

AI Daily Wellness Tip

Take 5 minutes between sprint tasks for deep diaphragmatic breathing to stabilize blood pressure.

This Week's Target Goal

Aim for 10,000 daily steps and complete 4 days of cardio workout routines.

AI-Generated Daily Routine

Generate New Routine

Click "Generate New Routine" for an AI-customized daily routine.



NUTRITION

AI Meal & Diet Plans

Personalized meal plans generated by AI based on your health profile.

CHOOSE YOUR DIET TYPE

Vegetarian

Vegan

Non-Veg

Diabetic

Weight Loss

Weight Gain

CALORIES
1600-1800 kcal kcal

PROTEIN
65-75g

WATER INTAKE
3 L

Breakfast

- Vegetable Oats Upma with peas and carrots (1 bowl)
- Mixed Moong Sprouts Salad with lemon and cucumber (1 cup)

Lunch

- 2 Whole Wheat Chapatis
- Paneer and Mixed Vegetable Sabzi (1 bowl)
- Low-fat Cucumber Raita (1 cup)

Dinner

- Moong Dal and Spinach Khichdi (1 bowl)
- Stir-fried Broccoli and French Beans

Snacks

- Roasted Chana / Chickpeas (1 handful)
- 1 Apple with 5-6 soaked Almonds

This plan is a general guideline based on your selected diet type — adjust portions based on your activity level, and consult a nutritionist for medical conditions.



➔ Added Features for Wellness

- Integrated AI wellness coach
- Added diet plans for the selected diet type based on user preference



Employee Dashboard



REPORTS

Health Reports & Timeline

Download health reports as PDF and view your complete history.



Download Health Report

Generates a PDF snapshot of your current wellness score, BMI, stress, sleep, and recommendations.



Download PDF



Old Reports Timeline

Every time your health profile is updated, a snapshot is saved here so you can track changes over time.

No history yet — update your health profile to start building a timeline.

Checkups & Emergency SOS

Schedule health checkups and trigger emergency SOS alerts.



Book a Check-up

DATE

dd-mm-yyyy



TYPE

Annual Health Check-up



NOTES (OPTIONAL)

+ Book Appointment

My Appointments

No check-ups booked yet.



Emergency SOS

Pressing this immediately alerts the admin/HR team with your emergency contact and known health info (blood group, allergies, conditions).



Added Features for Wellness

- Downloadable Health reports in PDF format.
- Schedule checkups and Emergency SOS trigger



Conclusion

- The project includes secure authentication, database integration, risk prediction, recommendation, sentimental analysis, performance analytics, and an AI wellness chatbot in one platform.
- It supports both employee and admin roles for managing wellness data and system features.
- These modules make the system more intelligent, interactive, and useful for future expansion.

**Thank you
so much**

